

Applications of Statistical Modeling:
Module #2, Lecture #4

Accounting for Complex Sample Designs When Fitting Structural Equation
Models

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Review: Complex Samples

- Recall from **Module 1** the notions of **complex sampling** and **design-based analysis**
- Use survey weights to compute design-unbiased estimates of parameters describing characteristics of a finite population
- Use either replicated variance estimation
 - requires replicate weights based on the sample designor Taylor Series Linearization approaches
 - requires variables describing stratum and cluster codesfor **design-unbiased** variance estimation

Overview of Today's Lecture

- How do we account for these sample design features when fitting structural equation models and performing latent class analysis?
- Review of the complex sample design features of the NESARC and available design variables
 - One cluster per stratum problem
- Revisit analyses of the NESARC data, now estimating SEMs that describe the finite target NESARC population, not just the sample

Overview of Today's Lecture, cont'd

- Consider a second example of fitting an SEM using the ESS data
- Discuss software options for fitting SEMs and performing LCA with complex sample survey data sets (Preview: not many!)
- Discuss how to describe these kinds of analyses in scientific publications

Conceptual Background

- **Critical Paper in this area:** Muthen and Satorra (1995); available on Canvas
- These authors described the conceptual and theoretical basis for fitting SEMs to complex sample survey data sets
- They established an estimation and inference framework for this setting: how should weighted estimates and design-adjusted SEs be computed?
- Stapleton (2006) provides a good overview of the conceptual background, and includes practical advice

Estimation of SEM Parameters

- “Conventional” SEM assumes that observations in a data set are independent and identically distributed (i.i.d., e.g., MVN)
- These assumptions seldom hold for complex samples; observations tend to be correlated
- Furthermore, ignoring **informative** survey weights (w.r.t. **variables** or **paths**) can lead to biased estimates of the parameters in an SEM, and misleading inferences

Estimation of SEM Parameters

- Recall the **fitting function** from **SEM Lecture #1**
- **Basic Idea (also applies to LCA):**
 - Compute **weighted** estimates of all of the distinct (sample) variances and covariances for variables in the model
 - When defining the overall likelihood function, apply the **final survey weight** to each individual's contribution
 - Given the weighted (population estimate!) of the overall likelihood function, find estimates of the parameters in the SEM that minimize the function
- See Stapleton (2006), page 36, for details

Estimation of SEM Parameters, cont'd

- Once parameters have been estimated, standard likelihood-based computations of standard errors (e.g., OIM, or observed information matrix in Stata) are **not** performed
 - These ignore the sample design features!
- Given the estimates, we then apply some form of Taylor Series Linearization or replication to compute design-based estimates of the variances
- Remember that variability in the weights also needs to be accounted for in estimation of standard errors!

Variance Estimation

- Muthen and Satorra (1995) develop a TSL solution for SEM models, and demonstrate its superiority over standard approaches to variance estimation when fitting SEMs to complex sample survey data
- Variance estimates for estimated parameters in SEMs can be extremely biased when assuming that the data arise from a simple random sample
- Stapleton (2006) performed a series of simulation studies to examine design effects when fitting SEMs to complex samples

Variance Estimation

- Collectively, this method (estimation based on a weighted likelihood function, followed by non-parametric design-based variance estimation) is also referred to as **pseudo-maximum likelihood (PML) estimation**; the best approach in Stapleton (2006)
- This is no different from other design-based methods that you will learn about!
- Replication methods **not** widely available (yet)

Variance Estimation, cont'd

- What about simply multiplying estimated standard errors (assuming SRS) by average design effects prepared by a data producer?
 - May be OK if weights are uninformative, but ignorance of weights can lead to biased estimates
- What about adjusting weights by average design effects, and normalizing weights?
 - What estimates should be used to compute the average design effects? Path coefficients? Variances?
 - Approach has worked well in previous studies, but Stapleton (2006) shows that it does not work well for SEM

Adjustments to Fit Statistics

- The standard Chi-square test statistic for goodness of fit (based on **weighted** parameter estimates) needs to be **adjusted** to account for the complex sampling features
- The basic idea is similar to the Rao-Scott adjustment for design effects you will learn about in other analysis courses (e.g., 701)
- See Stapleton (2006, page 37) for details

Adjustments to Fit Statistics, cont'd

- Other commonly used measures of fit can also be easily adapted to the complex sample setting, by replacing unweighted quantities used to compute the measures of fit with the corresponding weighted quantities
 - RMSEA
 - CFI
- Everything is just extended to the finite population inference framework!

Alternative Model-Based Approach

- Also discussed in Muthen and Satorra (1995)
- **Basic idea:** fit a structural equation model using random effects to account for clustering (similar to multilevel modeling)
- Muthen and Satorra refer to this as a “disaggregated” approach to the analysis
- This approach would be best if there are specific questions related to between-cluster variance in relationships of interest

Available Software

- What software procedures can be used to fit SEMs to complex sample survey data?
 - Mplus (Recommended)
 - The `lavaan.survey` package in R (Oberski, 2014; see article on Canvas; recently dropped from CRAN?)
 - PROC LCA (LCA only) or IVEware + CALIS in SAS
 - Stata: `svy: sem/gsem` after running `svyset`
 - Latent GOLD
 - LISREL (see Appendix example in Stapleton, 2006)
 - *Nothing else (as of this class)!*

Revisiting the NESARC Design

- Recall that a complex sample design was used for the NESARC, involving stratified multi-stage sampling of U.S. households
- Unfortunately, the design information provided in the public-use data set prohibits straightforward variance estimation!
- Why? Consider the output on the next slide from Stata's `svydes` command...

Available NESARC Design Variables

Survey: Describing stage 1 sampling units

```
pweight: w2weight  
VCE: linearized  
Single unit: missing  
Strata 1: w2stratum  
SU 1: w2psu  
FPC 1: <zero>
```

Stratum	#Units	#Obs	#Obs per Unit		
			min	mean	max
101	2	257	101	128.5	156
103	1*	131	131	131.0	131
104	1*	73	73	73.0	73
105	1*	50	50	50.0	50
106	1*	59	59	59.0	59
107	1*	25	25	25.0	25
108	1*	27	27	27.0	27
109	1*	20	20	20.0	20
110	1*	16	16	16.0	16
111	1*	20	20	20.0	20
201	2	32	14	16.0	18
203	1*	38	38	38.0	38

Note that there are several strata with only one cluster!

Dealing with “Singletons”

- We have a “singleton” problem, prohibiting the use of linearization
- Recall that TSL requires two clusters per stratum for variance estimation!
- We need to split singleton PSUs into two **random groups** (Muthen and Satorra, 1995)
 - Procedure does not bias variance estimates
- Within a PSU: even-numbered observations join one “cluster”, odd-numbered a second...

Dealing with “Singletons”

- **Note:** some random groups might be small, complicating subgroup analyses
- Another option would be to collapse “similar” strata, forming new pseudo-strata with multiple (2?) clusters present per pseudo-stratum
- But how do we know which strata are “similar”? May need to work with NESARC support staff...
- Some software provides “ad-hoc” options...e.g., see `help svyset` and `singleunit()` in Stata

Revisiting the NESARC LCA

- Given a fix for the variance estimation problem, let's revisit the LCA that we performed last week
- Recall that we would like to examine possible latent classes defined by heavy drinking and the nonmedical use of four classes of prescription drugs in the past year
- But now, we want to estimate the probability of falling into each class in the **target NESARC population**, using the survey weights!

Revisiting the NESARC LCA

- Revised Mplus Syntax:

```
TITLE:          NESARC LCA Example, Complex Sampling Features
DATA:           FILE = M:\nesarc_w1w2_red_wts.dat;
VARIABLE:       NAMES = idnum anysedy anytrqy anyopiy anystmy hd w2weight w2psu
                  w2stratum anyduda w2_psu_new;
AUXILIARY = idnum anyduda;
USEVARIABLES = anysedy anytrqy anyopiy anystmy hd w2weight
                  w2stratum w2_psu_new;
CATEGORICAL = anysedy anytrqy anyopiy anystmy hd;
WEIGHT = w2weight;
CLUSTER = w2_psu_new;
STRATIFICATION = w2stratum;
MISSING = anysedy (999) anytrqy (999) anyopiy (999) anystmy (999) hd (999);
          CLASSES = c(3);
ANALYSIS:       TYPE = MIXTURE;
                  TYPE = COMPLEX;
PLOT:           TYPE = PLOT3;
                  SERIES = anysedy anytrqy anyopiy anystmy hd(*);
OUTPUT:         TECH11; !TECH14 (BOOTSTRAP LRT) NOT AVAILABLE WHEN TYPE = COMPLEX
SAVEDATA:       FILE IS M:\cprobs.dat;
                  SAVE IS CPROB;
```

New cluster variable
containing random
groups for each stratum
with a “singleton” PSU

Revisiting the NESARC LCA

- Selected output (“SRS” results in parentheses)

Information Criteria

Akaike (AIC)	52232.822
Bayesian (BIC)	52376.525
Sample-Size Adjusted BIC	52322.500
$(n^* = (n + 2) / 24)$	

Chi-Square Test of Model Fit for the Binary and Ordered Categorical (Ordinal) Outcomes**

Pearson Chi-Square

Value	38.138 (32.784; Less evidence of a good fit)
Degrees of Freedom	13
P-Value	0.0003 (0.0018)

Likelihood Ratio Chi-Square

Value	36.551 (37.433)
Degrees of Freedom	13
P-Value	0.0005 (0.0004)

Revisiting the NESARC LCA

FINAL CLASS COUNTS AND PROPORTIONS FOR THE LATENT CLASS PATTERNS
BASED ON ESTIMATED POSTERIOR PROBABILITIES

Latent
Classes

1	1342.04871	0.03873 (0.03181)
2	302.56308	0.00873 (0.00885)
3	33008.38821	0.95254 (0.95935)

CLASSIFICATION QUALITY

Entropy	0.892 (0.904)
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Revisiting the NESARC LCA

RESULTS IN PROBABILITY SCALE

Latent Class 1

ANYSEDY

Category 1	0.919	0.054	17.162	0.000
Category 2	0.081	0.054	1.505	0.132 (NOTE: No longer evidence of this probability > 0!)

ANYTRQY

Category 1	0.936	0.055	17.102	0.000
Category 2	0.064	0.055	1.179	0.238 (SAME CHANGE: SEs Increase!)

ANYOPIY

Category 1	0.791	0.111	7.132	0.000
Category 2	0.209	0.111	1.879	0.060 (MARGINAL EVIDENCE)

ANYSTMY

Category 1	0.929	0.029	32.011	0.000
Category 2	0.071	0.029	2.439	0.015

HD

Category 1	0.201	0.096	2.095	0.036
Category 2	0.799	0.096	8.324	0.000 (STILL THE HEAVY DRINKING CLASS...)

Revisiting the NESARC LCA

VUONG-LO-MENDELL-RUBIN LIKELIHOOD RATIO TEST FOR 2 (H0) VERSUS 3 CLASSES

H0 Loglikelihood Value	-26142.789
2 Times the Loglikelihood Difference	86.756
Difference in the Number of Parameters	6
Mean	21.930
Standard Deviation	28.162
P-Value	0.0263

LO-MENDELL-RUBIN ADJUSTED LRT TEST

Value	85.395 (82.177)
P-Value	0.0279 (0.0000)

We still reject the null hypothesis of two classes, but the evidence is not as strong when accounting for the complex sampling features!

Revisiting the NESARC LCA

- Predicted plots look essentially the same, BUT:
 - Predicted probabilities have higher standard error associated with them, due to the stratified cluster sampling
- Use of weights did not make a big difference in the estimated probabilities (for classes or items); this will not always be the case!

An Example using the ESS Data

- We'll now consider a second example of SEM for complex samples, analyzing the ESS-Belgium data
- One of the latent factors will be defined by the following variables:
 - IMSMETN: The country should allow many/few immigrants of the same race / ethnicity as the majority (1 = Many, 4 = Few)
 - IMDFETN: “ “ of different race / ethnicity “ “
 - IMPCNTR: “ “ from poorer countries outside Europe
- **Possible latent factor:** general attitude toward immigration

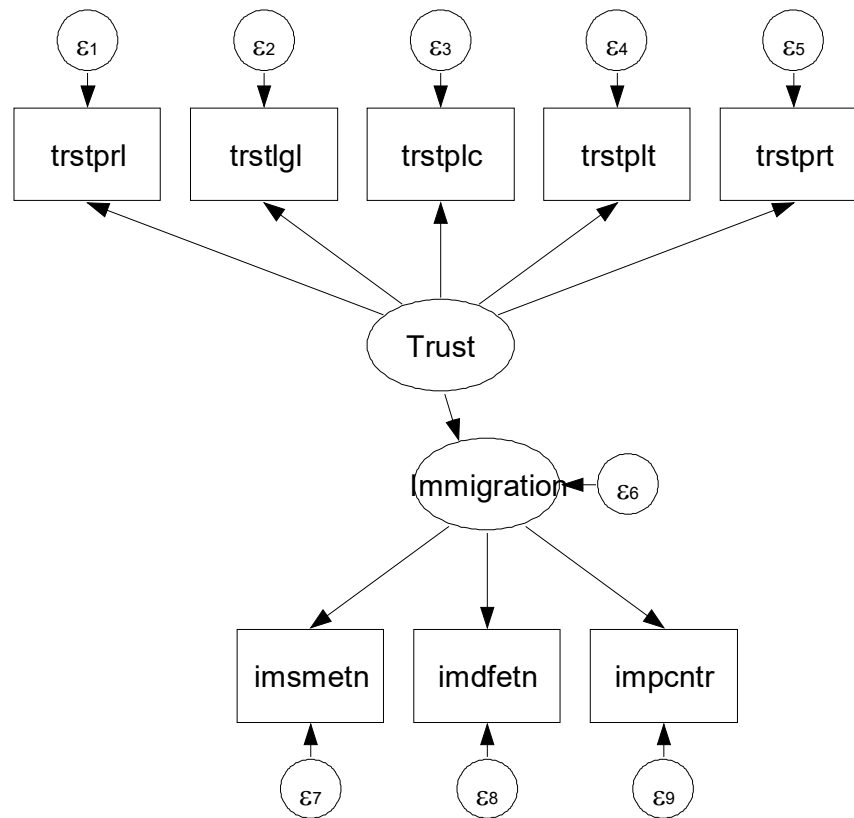
An Example using the ESS Data

- We will then predict the latent immigration factor with a second latent factor, indicated by five variables measuring **trust in authorities**:
 - TRSTPRL (parliament, range 0 to 10, 88 missing)
 - TRSTLGL (legal system)
 - TRSTPLC (police)
 - TRSTPLT (politicians)
 - TRSTPRT (political parties)

An Example using the ESS Data

- Is this a complex sample?
 - Typically, one interviewer is assigned to work in one PSU (Interviewer $\Leftrightarrow$ PSU)
 - We will use the inverse of the response rate within a PSU (interviewer) as the final weight, for illustration purposes
- Note that values of 8 on the immigration variables represent missing data; always important to check when prepping for Mplus!

SEM for the ESS Example



ESS Example: Mplus Syntax

TITLE: ESS CFA Example, Complex Sampling Features

DATA: FILE = M:\ess_belgium_red.dat;

VARIABLE: NAMES = intnum trst1 trst2 trst3 trst4 trst5
imm1 imm2 imm3 weight;
USEVARIABLES = intnum trst1 trst2 trst3 trst4 trst5
imm1 imm2 imm3 weight;
WEIGHT = weight;
CLUSTER = intnum;
MISSING = trst1 (88) trst2 (88) trst3 (88) trst4 (88) trst5 (88)
imm1 (8) imm2 (8) imm3 (8);

ANALYSIS: **TYPE = COMPLEX;**

MODEL: F1 BY imm1 imm2 imm3;
F2 BY trst1 trst2 trst3 trst4 trst5;
F1 ON F2;

OUTPUT: STANDARDIZED SAMPSTAT;

ESS Example: R Syntax

```
library(survey)
library(lavaan)
library(lavaan.survey)

# Create a survey design object specifying the clusters and weights
svydes <- svydesign(ids=~intnum, prob=~weight, data=ess.belgium)

# Fit model using lavaan()
belg.model <- ` model specification goes here in single quotes `
model.fit <- lavaan(belg.model, data=ess.belgium, ...)

# Given a model fit object generated by lavaan(), submit it as the
# lavaan.fit argument to lavaan.survey()
survey.fit <- lavaan.survey(lavaan.fit=model.fit, survey.design=svydes)
summary(survey.fit)
```

ESS Example: Mplus Output

RMSEA (Root Mean Square Error Of Approximation)

Estimate	0.103
90 Percent C.I.	0.094 0.112
Probability RMSEA <= .05	0.000 (fit not great...)

CFI/TLI

CFI	0.917
TLI	0.878

ESS Example: Mplus Output

MODEL RESULTS

		Estimate	S.E.	Est./S.E.	Two-Tailed P-Value
F1	BY				
	IMM1	1.000	0.000	999.000	999.000
	IMM2	1.315	0.053	24.621	0.000
	IMM3	1.122	0.047	23.938	0.000
F2	BY				
	TRST1	1.000	0.000	999.000	999.000
	TRST2	0.877	0.032	27.088	0.000
	TRST3	0.667	0.033	20.129	0.000
	TRST4	1.234	0.043	28.907	0.000
	TRST5	1.205	0.045	26.807	0.000
F1	ON				
F2		-0.099	0.013	-7.796	0.000

With higher trust,
more immigration
desired!

ESS Example: Mplus Output

R-SQUARE (Note possible reasons for the poor fit!)

Observed Variable	Estimate	S.E.	Est./S.E.	Two-Tailed P-Value
TRST1	0.528	0.027	19.784	0.000
TRST2	0.374	0.027	13.664	0.000
TRST3	0.270	0.025	10.944	0.000
TRST4	0.847	0.019	43.838	0.000
TRST5	0.817	0.019	42.526	0.000
IMM1	0.590	0.033	17.783	0.000
IMM2	0.872	0.026	34.060	0.000
IMM3	0.685	0.032	21.597	0.000

Latent Variable	Estimate	S.E.	Est./S.E.	Two-Tailed P-Value
F1	0.068	0.017	3.966	0.000

Some Final Thoughts

- The same output is obtained using R / Stata / SAS / etc., but Mplus does a better job of providing common design-adjusted measures of fit
- The aforementioned software can fit design-adjusted SEMs to continuous and categorical variables
- Important to assess whether weights are informative about key estimates; if they are not, using them in estimation may inflate standard errors artificially (just as in simpler regression models!)